Modeling Avalanche risk with python: ava_risk

Jacob, Ethan, Bryn

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Abstract

This paper introduces ava_risk: a python library that allows for the modeling of the risk of an avalanche on a slope.

1 Introduction

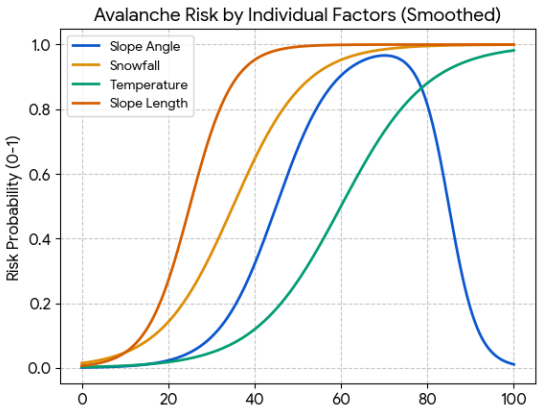
There are four major factors we are taking into account when judging the probability of an avalanche:

1. Slope angle - steepness of the terrain
2. Temperature - air temperature around the given slope
3. Fresh snowfall - average depth of snowfall fallen within the last 24 hours
4. Slope length - length from bottom to top of the given slope

The *avariskpy* library plans to model the risk of an avalanche on a given slope by taking into account the environment around the the slope. At its core, the model represents avalanche risk as a function of interacting environmental variables. Each factor is treated as an input that contributes. An example of this is:
>>> inport avalanche
>>>slope_angle=avariskpy.compute_slope_angle(368,100,1000)
>>> compute_slope_angle_risk(slope_angle)
>>> 0.2
As shown the model uses simulations to assess the variables and estimate the resulting avalanche risk. The simulation illustrates how changes in environmental conditions affect overall risk.

2 Statement of need

This type of analysis is very complex, for example [1] shows some academic work aimed at assessing the model around avalanche forecasting. This highlights the mathematical modeling of the assessed variables.
In [1] the paper dives into the importance of fresh snowfall and the slope angle in predicting an avalanche and uses mathematic models and images to display this. This is essentiated by the [2] Avalanche danger scale; this library uses data from the avalanche danger scale and the scale for avalanche prediction.
The *ava_risk* library is the first Python library (to the Authors knowledge) that can be used to predicted an avalanche on any given slope and can be conjunction with other scientific tools.



The graph shows plotted data from [2] for probability of an avalanche for each factor adjusted to standardized units.

3 Conclusion

This paper gives a description of *ava_risk*: a library built for mathematically predicting the probability of avalanche. Currently the library has limitations, such as the assumption that the slope is flat, but the object orientated nature of the library can be used to extend it out to different surfaces
ava_risk is modularised, tested and documented.

References

[1] Cristina Pérez-Guillén. *Assessing the performance and explainability of an avalanche danger forecast model.* Natural Hazards Earth System Sciences, 2025.

[2] European Avalanche Warning Services. Avalanche danger scale,. *(EWAS)*, 1993 –present.